

Lecture 4

Matrices of Linear Maps; Change of Basis

Throughout, bases are ordered and $\mathcal{M}(T)$ denotes the matrix of T with respect to the bases in use. Tags indicate the flavour of each problem; a ★ marks a harder one.

Core tutorial route

Work **4.1, 4.7, 4.11, 4.14, 4.15, 4.16, 4.17, and 4.20** in that order. This route covers building the matrix of a map, commuting operators and their matrices, testing invertibility, the dimension of a map space, the equivalence of injectivity, surjectivity and invertibility, computing a matrix in a new basis, converting coordinates, and the proof that trace and determinant are invariant.

Additional practice: 4.3–4.4, 4.8–4.10, 4.12–4.13, 4.18–4.19. **Bridge:** 4.2, 4.5, 4.6 (differentiating polynomials). **Extension:** 4.21–4.23.

Matrices of maps

Exercise 4.1 (computational) Let $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ be $T(x, y, z) = (x + 2y - z, 3x + z)$. Write down the matrix $\mathcal{M}(T)$ with respect to the standard bases and use it to compute $[T(1, 1, 1)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{pmatrix}$; $[T(1, 1, 1)] = (2, 4)$.

Exercise 4.2 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be the differentiation operator. In the standard basis $1, t, t^2$ compute the matrices $\mathcal{M}(D)$ and $\mathcal{M}(D)^2$. Which operator has the matrix $\mathcal{M}(D)^2$?

Answer. $\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$, $\mathcal{M}(D)^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D^2)$.

Exercise 4.3 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ be $T(x, y) = (2x - y, x + 3y)$. Write down the matrix $\mathcal{M}(T)$ in the standard basis and use it to compute $[T(1, 2)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$; $[T(1, 2)] = (0, 7)$.

Exercise 4.4 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ be $T(x, y) = (x - y, 2x + y, 4y)$. Write down the matrix $\mathcal{M}(T)$ with respect to the standard bases and use it to compute $[T(3, -1)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 0 & 4 \end{pmatrix}$; $[T(3, -1)] = (4, 5, -4)$.

Exercise 4.5 (computational) Let $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be $(Tp)(t) = p(t) + t p'(t)$. In the standard basis $1, t, t^2$ write down the matrix $\mathcal{M}(T)$ and use it to compute $T(2 - t + 3t^2)$.

Answer. $\mathcal{M}(T) = \text{diag}(1, 2, 3)$; $T(2 - t + 3t^2) = 2 - 2t + 9t^2$.

Exercise 4.6 (computational) Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be the differentiation operator. Its matrix in the standard basis $1, t, t^2, t^3$ is

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Compute $\mathcal{M}(D)^3$ and find the smallest k such that $D^k = 0$.

Answer. $\mathcal{M}(D)^3 = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$; the smallest k is $k = 4$.

Exercise 4.7 (proof) Let $\dim V = n$, and let $S, T \in \mathcal{L}(V)$ be operators whose matrices in a fixed basis are $\mathcal{M}(S), \mathcal{M}(T)$. Prove that the operators commute, $ST = TS$, if and only if their matrices commute, $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S)$.

Hint. The theorem “A matrix acts on coordinates”, applied twice, gives $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$. Then use the fact that the map $\mathcal{M}$ is injective: operators are equal if and only if their matrices are equal.

Exercise 4.8 (proof) Fix bases of V and W . Prove that for any $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, $\mathcal{M}(aS + T) = a\mathcal{M}(S) + \mathcal{M}(T)$.

Hint. Compare the matrices column by column. The k -th column of $\mathcal{M}(aS + T)$ is the coordinate vector of $(aS + T)v_k$, and the coordinate map is linear.

Invertibility and isomorphism

Exercise 4.9 (computational) Decide whether the operator $T(x, y) = (2x + y, 4x + 2y)$ on $\mathbb{R}^2$ is invertible. If it is, find T^{-1} . If not, find a nonzero vector in the kernel $\ker T$.

Answer. Not invertible ($\det = 0$); $(1, -2) \in \ker T$.

Exercise 4.10 (computational) Show that the polynomials $1, 1 - t, (1 - t)^2$ form a basis of $\mathcal{P}_2(\mathbb{R})$. Transport the question to $\mathbb{R}^3$.

Answer. The coordinate vectors $(1, 0, 0), (1, -1, 0), (1, -2, 1)$ are linearly independent ($\det = -1$), so the polynomials form a basis.

Exercise 4.11 (computational) Decide whether the operator $T(x, y) = (3x + 5y, x + 2y)$ on $\mathbb{R}^2$ is invertible. If it is, write down a formula for T^{-1} .

Answer. Invertible ($\det = 1$); $T^{-1}(x, y) = (2x - 5y, -x + 3y)$.

Exercise 4.12 (computational) Decide whether the operator $T(x, y, z) = (x + 2y + 3z, 2x + 4y + 6z, x + y + z)$ on $\mathbb{R}^3$ is invertible. If not, find a nonzero vector in the kernel $\ker T$.

Answer. Not invertible ($\det = 0$); $(1, -2, 1) \in \ker T$.

Exercise 4.13 (computational) Show that the polynomials $1 + t^2, t + t^3, 1 + t, t^2$ form a basis of $\mathcal{P}_3(\mathbb{R})$. Transport the question to $\mathbb{R}^4$.

Answer. The determinant of the coordinate vectors is $1 \neq 0$, so the polynomials are linearly independent and form a basis.

Exercise 4.14 (proof) Prove that if V and W are finite-dimensional, then $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Hint. Show that the map $\mathcal{M}: \mathcal{L}(V, W) \rightarrow \mathbb{F}^{m \times n}$ is an isomorphism (a matrix prescribes exactly the images of the basis vectors), and compute the dimension of the matrix space.

Exercise 4.15 (proof) Let V be finite-dimensional and let $T \in \mathcal{L}(V)$. Prove that the following are equivalent: (i) T is injective; (ii) T is surjective; (iii) T is invertible.

Hint. Use rank-nullity: $\dim \ker T + \dim \operatorname{im} T = \dim V$.

Change of basis and similarity

Exercise 4.16 (computational) Let $T(x, y) = (x, -y)$ be the reflection in the x -axis on $\mathbb{R}^2$. Its matrix in the standard basis is $A = \text{diag}(1, -1)$. Using the formula $A' = P^{-1}AP$, find the matrix of this reflection in the basis $\mathcal{B}' = ((1, 1), (1, -1))$.

Answer. $A' = P^{-1}AP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Exercise 4.17 (computational) Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$ and $\mathcal{B}' = ((2, 1), (1, 1))$. The coordinate vector of a vector v in the new basis is $[v]_{\mathcal{B}'} = (1, 2)$. Find $[v]_{\mathcal{B}}$ (that is, the vector v itself). Also find the coordinate vector $[w]_{\mathcal{B}'}$ of the vector $w = (3, 0)$.

Answer. $[v]_{\mathcal{B}} = (4, 3)$; $[w]_{\mathcal{B}'} = (3, -3)$.

Exercise 4.18 (computational) The matrix of an operator $T \in \mathcal{L}(\mathbb{R}^2)$ in the standard basis is $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Using the formula $A' = P^{-1}AP$, find its matrix in the basis $\mathcal{B}' = ((1, 0), (1, 1))$.

Answer. $A' = \text{diag}(2, 3)$.

Exercise 4.19 (computational) Using only trace and determinant, decide whether the matrices in each pair can be similar:

$$(a) \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}; \quad (b) \quad \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Answer. (a) No: the traces differ, $5 \neq 4$. (b) The invariants do not rule similarity out: both matrices have trace 4 and determinant 4.

Exercise 4.20 (proof) Prove that similar matrices have the same determinant and the same trace.

Hint. For the determinant, use $\det(XY) = \det X \det Y$. For the trace, use the identity $\text{tr}(XY) = \text{tr}(YX)$, grouping the product $P^{-1}AP$ as $(P^{-1}A)P$.

Exercise 4.21 (★) Suppose the matrix A is similar to the matrix B , and B is similar to C . Prove that A is similar to C . Then prove that similarity is an equivalence relation on the set of $n \times n$ matrices.

Hint. For transitivity, multiply the two matrices that give the similarities and simplify the expression $(PQ)^{-1}A(PQ)$.

Exercise 4.22 (★) Suppose A and B are $n \times n$ matrices and A is invertible. Prove that the matrices AB and BA are similar, and deduce that $\text{tr}(AB) = \text{tr}(BA)$.

Hint. You need a matrix P such that $P^{-1}(AB)P = BA$. Since A is invertible, try $P = A$.

Exercise 4.23 (★) Give two 2×2 matrices whose traces and determinants agree, but which are *not* similar. Prove that they are not similar.

Hint. Compare the matrix $3I$ with a non-diagonal matrix that has the same trace and determinant. What is the product $P^{-1}(3I)P$?

Answer. For example, $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ and $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$: both have trace 6 and determinant 9, but they are not similar.